

Karthigeyan Ganesh Shankar

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WORK EXPERIENCE

Lead Autonomy Engineer

April 2025 – May 2026

Ati Motors, India ([Website](#))([Blog](#))

Bengaluru, India

- Pioneered organization wide use (Solutions, Validation and Dev teams) of optimized **simulation-first** development and testing. Made the workflow available for Cloud, on-prem and edge environments
- Initiated the map-free navigation models which are generalized and SOTA (**NoMad**, **ViNT**), proactively identifying challenges before porting it to production and tweaking for Ati's vehicle archetypes
- Steered the evaluation of novel, SOTA 3D reconstruction (**VGGT**, **Mast3R**, **Splatfacto**) and VSLAM methods (**Mast3R-SLAM**, **Droid-SLAM**) to migrate towards vision based navigation and representation
- **Software/Language Used:** [Nvidia Isaac Sim](#), [Nvidia Warp](#), [Pytorch](#), [Blender](#), [MeshLab](#), [Cloudcompare](#)

Senior Autonomy Engineer

April 2024 – April 2025

Ati Motors, India ([Website](#))

Bengaluru, India

- Spearheading a team which is developing an one button press simulation workflow by aligning the current software stack with **Nvidia's Isaac Sim** application. The effort entails setting up **sim2real** environment and emulating existing control, navigation and perception stack of the bots into simulation
- Solely responsible for evaluating different OTS carrier boards (**Jetson**) that controls the entire bot by semi-automating the bring-up process and saving considerable dev and validation time
- Responsible towards building business critical, **production ready architecture** and implementation framework for both the bot and fleet manager application and making factual recommendations
- **Software/Language Used:** [Nvidia Isaac Sim](#), [Kubernetes](#), [Bazel](#), [Python](#), [Make](#), [Unix](#), [Embedded C](#)

Autonomy Engineer

Sept 2022 – March 2024

Ati Motors, India ([Website](#))

Bengaluru, India

- Revived and optimized the software build systems to catalyse the build process by **40 %** enabling the team to deploy features/bug fixes at speed with utmost reliability
- Developed the software-in-loop (CI/CD) frameworks to **automate** the software release process and reduce process bottlenecks with **bash scripting** and offline testing exigencies
- Enabled the team to use designed systems across all available software architectures increasing their productivity and throughput by **30%** by provided necessary training to use them well
- Supported in code development, robotic **deployment** and gained a working understanding of the industrial market and product development cycle of different products as part of the portfolio
- **Software/Language Used:** [Kubernetes](#), [Bazel](#), [Poetry](#), [Docker](#), [C++](#), [Python](#), [Make](#), [Unix](#), [Bash](#)

Research Assistant

Oct 2018 – Mar 2019

Institute for Aerodynamics and Fluid Mechanics, RWTH Aachen ([Website](#))

Aachen, Germany

- Devised massively parallel simulations in C++ for **1 million cells through 24 cores for 100+ core hours** to determine the optimal configuration for a laminar pre-mixed flame propagation. Subsequently, **automated** the visualization pipeline through Python
- Investigated the configuration through grid convergence study to determine minimal number of cells required to capture the physics effectively hence optimizing the simulation core hours
- **Software/Language Used:** [C++](#), [Python](#), [ParaView](#), [Unix](#), [Bash](#), [OpenMPI](#)

Research Intern

June 2016 – July 2016

Heat Transfer Lab, Indian Institute of Technology Guwahati ([Website](#))

Guwahati, India

- Collated projects pertaining to heat recovery and utilization systems. Introduced to both experimental and computational methodologies for porous media storage, solar dryer and latent heat storage systems
- Familiarized with COMSOL for modelling porous media with **100 pores** in asymmetric locations.
- Assisted experimental correspondent with a solar air dryer by recording and cleaning experimental data obtained through LabView
- **Software/Language Used:** [COMSOL](#), [Python](#), [LabView](#)

EDUCATION

Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen

Master of Sciences in Simulation Sciences

Oct 2017 – Mar 2020

Aachen, Germany

SRM University

Bachelor of Technology in Mechanical Engineering

May 2013 – May 2017

Chennai, India

SKILLS

Programming Languages: Python, C++, MATLAB, Bash, Rust (Beginner)

Tools: Dofin, ScoreP, $\LaTeX$, Gmsh, ParaView, NumPy, SciPy, Docker, Poetry, Blender, Nvidia Isaac Sim, MeshLab, CloudCompare, Bazel, Makefile, Kubernetes, Warp

Languages: English (Fluent), French (Intermediate), German (Beginner), Hindi (Intermediate), Tamil (Native)

OTHER EXPERIENCES

Repository homepage: [GitHub](#)

Master Thesis: Kinetic Model on Unemployment, RWTH Aachen

- Conceptualized, developed and analysed **mesoscopic models** to spatially and temporally characterize unemployment
- Performed involved **Monte-Carlo simulations** for case scenarios of increasing complexity
- Simulated and characterized the interactive, time-variant complex system and analytically concluded the upper bounds
- Advanced a niche area in transport equations with **minuscule** prior literature and works

Unstructured Finite Element Solver, RWTH Aachen

- Programmed a solver to operate on unstructured meshes using **Python (FeNiCS)**. Gmsh was used to generate unstructured meshes and simulation results were viewed using ParaView. In order to optimize the process, **convergence** studies was held
- Analysed the sensitivity of temperature profiles through **automatic differentiation** (dofin-adjoint) and optimized it for minimal computation time and complexity
- Enhanced the code further through **parallelization** and smart implementation to calculate the Jacobians quickly

Convection Coupled Simulation, RWTH Aachen

- Developed an enthalpy method for convection-coupled phase change simulation with Python to **reduce the complexity** of the simulation. Reformulated the governing equations and implemented its weak form in Python (FeNiCs)
- Validated the results against a benchmark (Stefan problem) by solving a **non-linear problem** through Newtons method

Bachelor Thesis: Fatigue Characteristics of Functionally Graded Materials, SRM University

- Supervised manufacturing of **Aluminium (L24)** with different filler materials (titanium oxide, silicon carbide, aluminium oxide) at **under-engineered locations** based on their stress profiles
- Administered and performed experimental analysis to obtain **natural frequency** of the manufactured material using a tri-axial accelerometer and subsequently fatigue frequency was calculated with respect to the frequency of Aluminium

Thermal Energy Storage System, SRM University

- Researched and designed a thermal energy storage system using **phase change** material and metal filings
- Compared and calculated the efficiency and heat capacity against conventional heat storage systems and improved them by **20%**